

Key Field Skills

- **EASA Drone Pilot License (09/2024)**
- GPS / field topographic surveys

GIS

- QGIS
- ArcGIS Pro
- PostGIS

Programming

- Python, SQL
- JavaScript, R, MATLAB

Analysis & AI

- Machine Learning / Deep Learning
- Satellite remote sensing
- Spatial analysis
- Environmental modeling
- Cartography

Languages

French — Native
English — C1
German — B2

Certificate

Remote Pilot Certificate, EASA (09/2024)

Links

LinkedIn
GitHub

Interests

Hiking · Nature · Cinema · Cooking · Music

Mohamed Bouzenad

Environmental & Spatial Data Analyst

Remote Sensing & AI Applied to Alpine Natural Hazards

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Profile

Data analyst specialized in remote sensing and deep learning applied to alpine natural hazards. Master's degree in Environmental Science (UNIL), with a thesis on Sentinel-2 image super-resolution via deep learning and two field projects on rock glacier monitoring and landslide susceptibility modeling. EASA drone pilot license.

Technical & Analytical Projects

Sentinel-2 Satellite Image Super-Resolution via Deep Learning

Master's Thesis — UNIL, supervised by Prof. Grégoire Mariéthoz | 2025 – 2026

- Designed a **Python** super-resolution model with a ×3 factor (resolution increased from 1 to 9 pixels).
- Pre-processing of multi-band satellite data and quantitative performance evaluation using **PSNR** and **SSIM**.

Diachronic Monitoring of the Lac des Vaux Rock Glacier

Mont-Gelé, Verbier | 2023 – 2024

- Displacement measurements via **DGNSS** on 47 boulders, producing a flow velocity map in **QGIS**.
- Drone photogrammetry (**Pix4D Mapper**), generating digital elevation models at 6 cm resolution and computing a Difference of DEMs (DoD).
- Deployment and analysis of 13 thermal sensors (**GST**) and an electrical resistivity tomography profile (**ERT**, 48 electrodes).
- Results: active displacements of up to approximately 1 m/year, active permafrost identified between 2690 and 2715 m altitude.

Landslide Susceptibility Modeling

Ubaye Valley, France | 2024 – 2025

- Modeling via **logistic regression** under **Python/QGIS**, based on topographic data (**DEM**) and multi-criteria analysis.

Education

MSc in Environmental Science — Earth Surface Processes in Mountain Environments, University of Lausanne (UNIL), 09/2023 – 01/2026

Bachelor's degree — Geosciences and Environment, University of Lausanne (UNIL), 09/2019 – 06/2021

Erasmus Exchange — Geography and Geology, University of Montreal, Canada, 08/2021 – 06/2022

Other Experience

Secretary — Pediatric Emergency Department, Lausanne Children's Hospital, 2024

Camp Counselor — Centre aéré des Allévays, Nyon, 2022 – 2024

References

Grégoire Mariéthoz — PhD Supervisor, Professor, Institute of Earth Surface Dynamics, UNIL — gregoire.mariethoz@unil.ch

Lucas Schmutz — PhD Candidate, UNIL — lucas.schmutz@unil.ch